

Table 1. False-premise detection across models and datasets

Method	Qwen2.5 7B				GPT-6 Luna				Gemma 4 12B				Qwen3.8 27B OFF				Qwen3.8 27B ON			
	Cancer-Myth		CREPE		Cancer-Myth		CREPE		Cancer-Myth		CREPE		Cancer-Myth		CREPE		Cancer-Myth		CREPE	
	TPR ↑	FPR ↓	TPR ↑	FPR ↓	TPR ↑	FPR ↓	TPR ↑	FPR ↓	TPR ↑	FPR ↓	TPR ↑	FPR ↓	TPR ↑	FPR ↓	TPR ↑	FPR ↓	TPR ↑	FPR ↓	TPR ↑	FPR ↓
TF-IDF text	Shared across models — Cancer-Myth: TPR ↑ 86.1%, FPR ↓ 51.0%; CREPE: TPR ↑ —, FPR ↓ —																			
Direct gate	7.0	10.7	—	—	—	—	—	—	40.5	16.1	—	—	26.1	16.1	—	—	44.1	21.5	—	—
CoT gate	29.5	16.1	—	—	—	—	—	—	51.3	32.9	—	—	57.5	35.6	—	—	61.4	38.3	—	—
Probe gate (hidden states)	75.0	47.7	—	—	n/a	n/a	n/a	n/a	—	—	—	—	—	—	—	—	—	—	—	—
GEPA-gate	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Stored results; —: not measured/comparable; n/a: outside the defined implementation. Percentages, one run.

H: historical full-corpus Cancer-Myth; valid denominators and protocol differences in ledger.

TPR ↑: FPQ detection; FPR ↓: false positives on normal questions.

Probes require hidden states; no native Luna probe. TF-IDF is model-independent.

Default crossfit probe/TF-IDF decisions; these are not matched-FPR thresholds.

Ours is response-only and is excluded. GEPA-gate denotes detector-prompt optimization in both tables.